Towards **Reliable Machine Learning** for **Visual Analytics**

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Reliable **Visual Analytics:**

The knowledge or decisions made by **analysts, or users,** should precisely reflect the underlying data.

I tackle **prevalent assumptions** that makes
ML applications in visual analyatics unreliable.

I **design** and **democratize** diverse techniques for the purpose

I focus on **dimensionality reduction** and **clustering**,
two ML algorithms that are widely used for visual analytics

Reliable Algorithms

- UMATO (**VIS 2022**)

Reliable Interactions

- CLAMS (**VIS 2023**)
  Best papers honorable mention (top 5%)
- Distortion-aware brushing (**wip**)
  under review (**TVCG**)

Reliable Evaluations

- Steadiness & Cohesiveness (**VIS 2021**)
- Classes are not Clusters (**VIS 2023**)
- Adjusted IVMs (**wip**)
  under review (**TPAMI**)

Democratizing the Findings

- ZADU (**VIS 2023**)
- Structural Complexity (**wip**)
  under review (**VIS 2024**)
- Why not be reliable? (**wip**)

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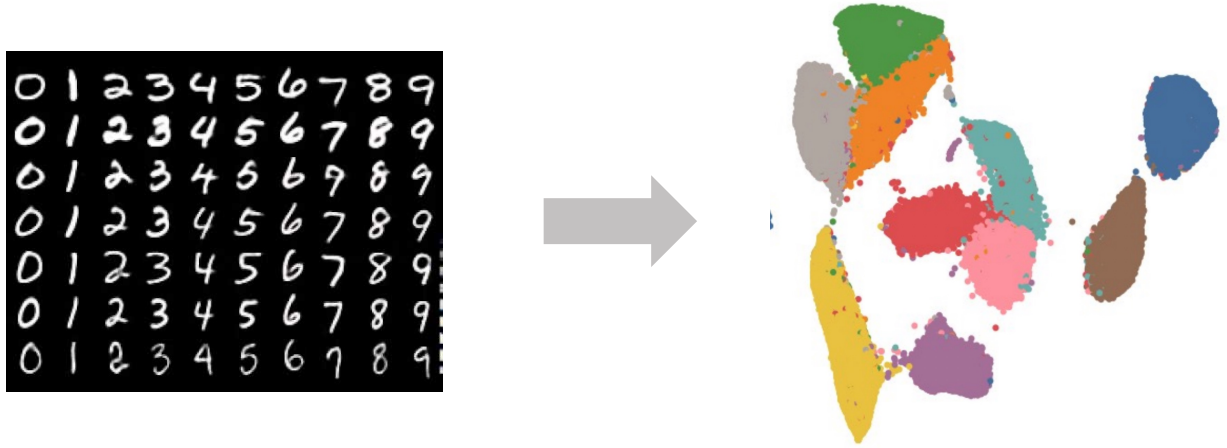


Reliable Algorithms

UMATO IEEE VIS 2022

UMAP is widely used for **cluster analysis**

However, the technique actually focus on **neighborhood structure**, thus using it for cluster analysis is unreliable



UMATO IEEE VIS 2022

I designed **UMATO,** a dimensionality reduction technique that reliably depicts both **neighborhood** and **cluster** structures



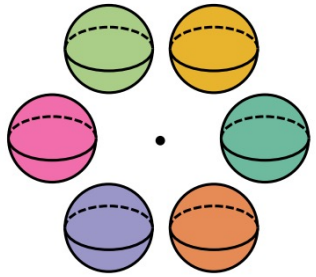




Reliable Evaluations

Class Labels are widely used as well-separated **ground truth cluster** in evaluating **dimensionality reduction** and **clustering**

However, **class labels** often do not form **clear clusters**, making evaluations using the labels **less reliable**



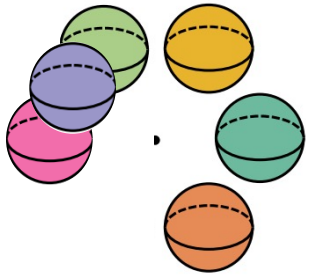
Class labels



Good embedding



Bad embedding



Class labels

!= Ground Truth Clusters





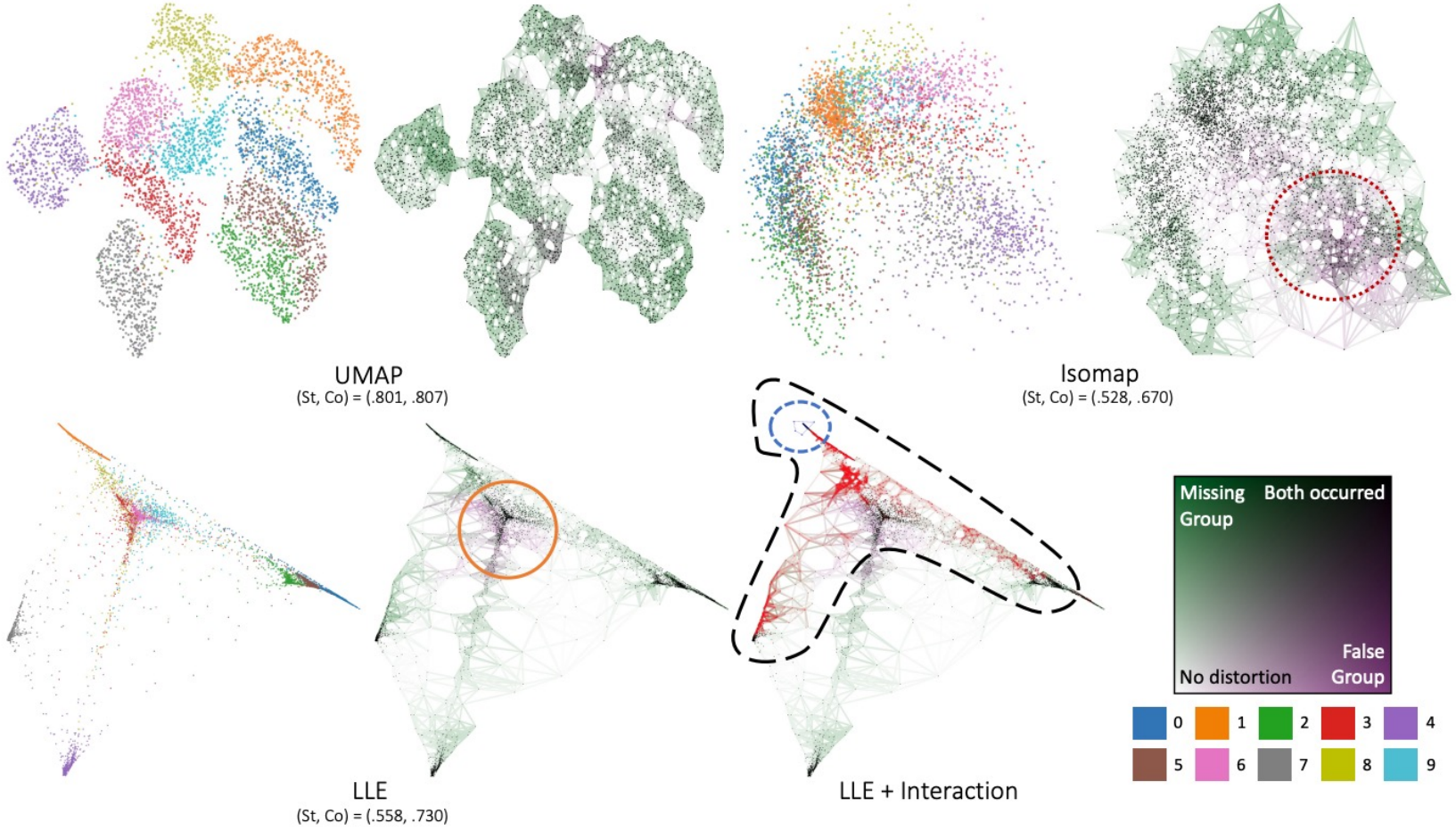
Good embedding??





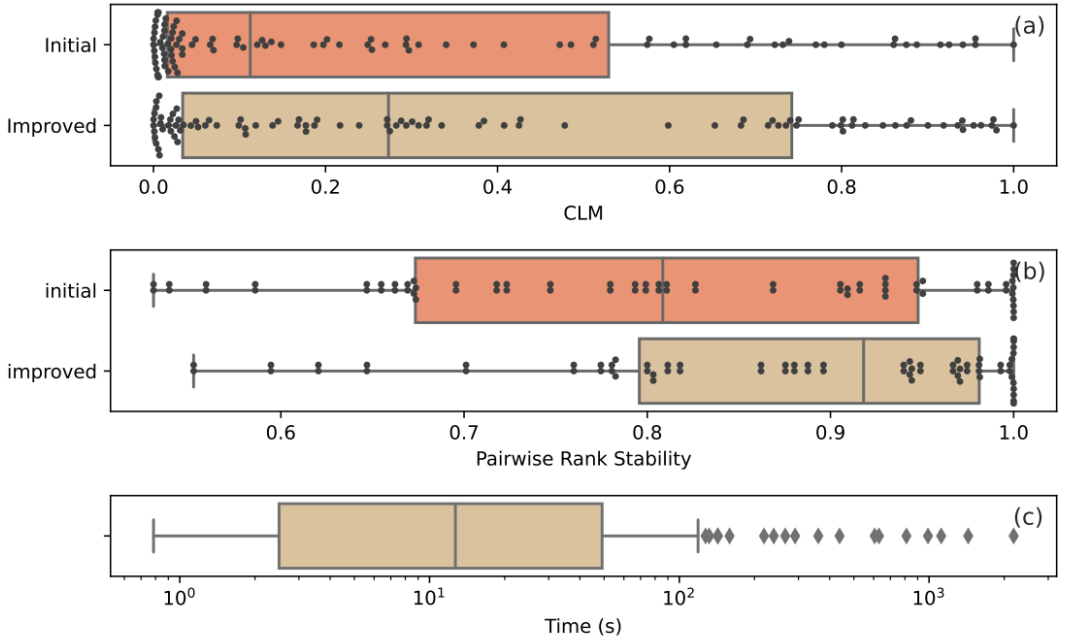
Bad embedding??

Steadiness & Cohesiveness IEEE VIS 2021



Steadiness & Cohesiveness IEEE VIS 2021

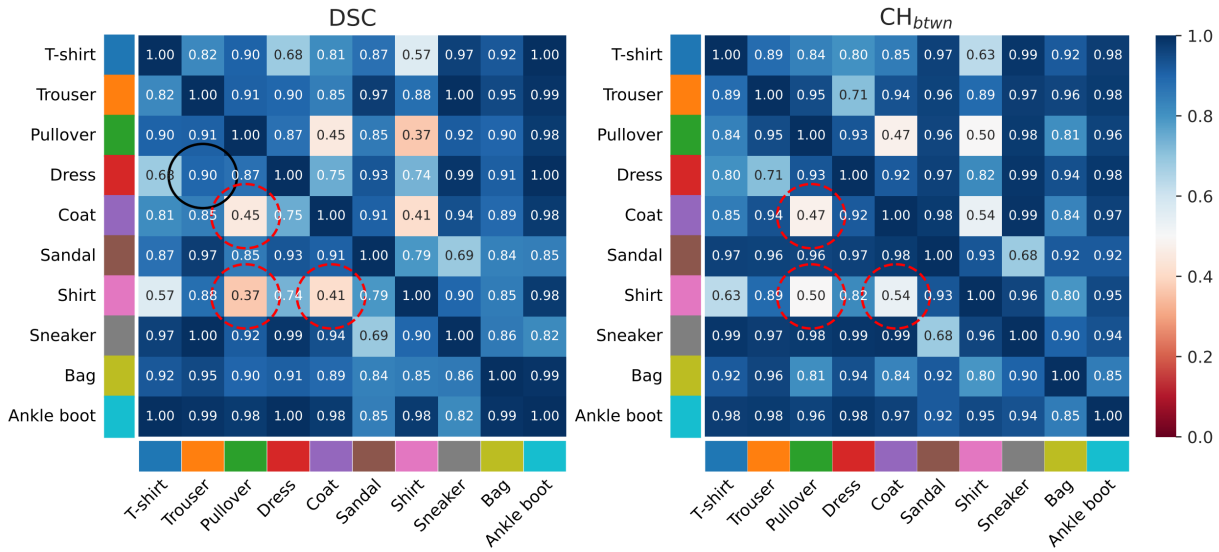
Adjusted IVMs *under review*, IEEE TPAMI



Steadiness & Cohesiveness IEEE VIS 2021

Adjusted IVMs *under review*, IEEE TPAMI

Classes are not Clusters IEEE VIS 2023



Steadiness & Cohesiveness IEEE VIS 2021

Let's **avoid using class labels!!**
Instead, let's use **clustering algorithms** to examine cluster structure

Adjusted IVMs *under review*, IEEE TPAMI

Let's check the separability of class labels
and **only use well-separated classes** as ground truth clusters

Classes are not Clusters IEEE VIS 2023

Let's check the separability of class labels
and use the **separability itself as ground truth**

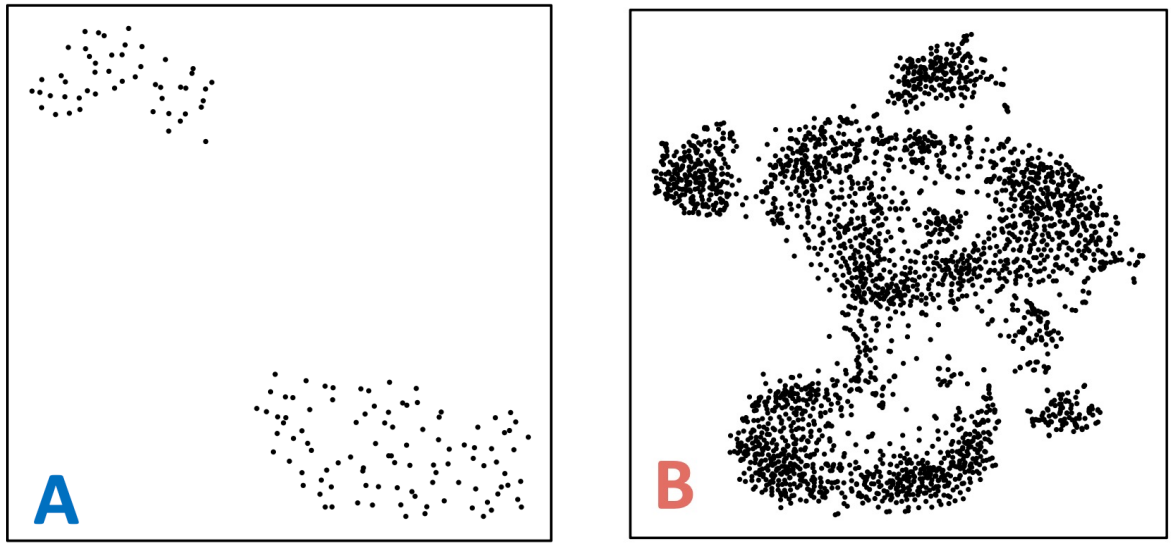


Reliable Interactions

CLAMS IEEE VIS 2023 *Best papers honorable mention (top 5%)*

Inconsistency between analysts can make visual analytics unreliable

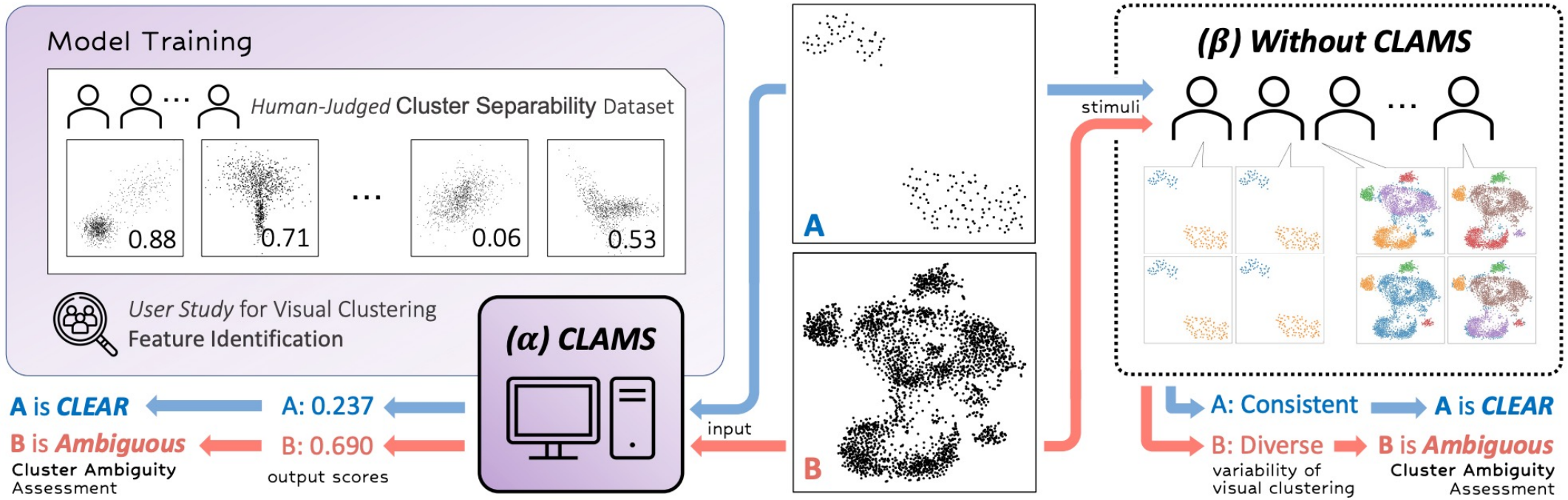
Ambiguous visualizations incurs such inconsistency



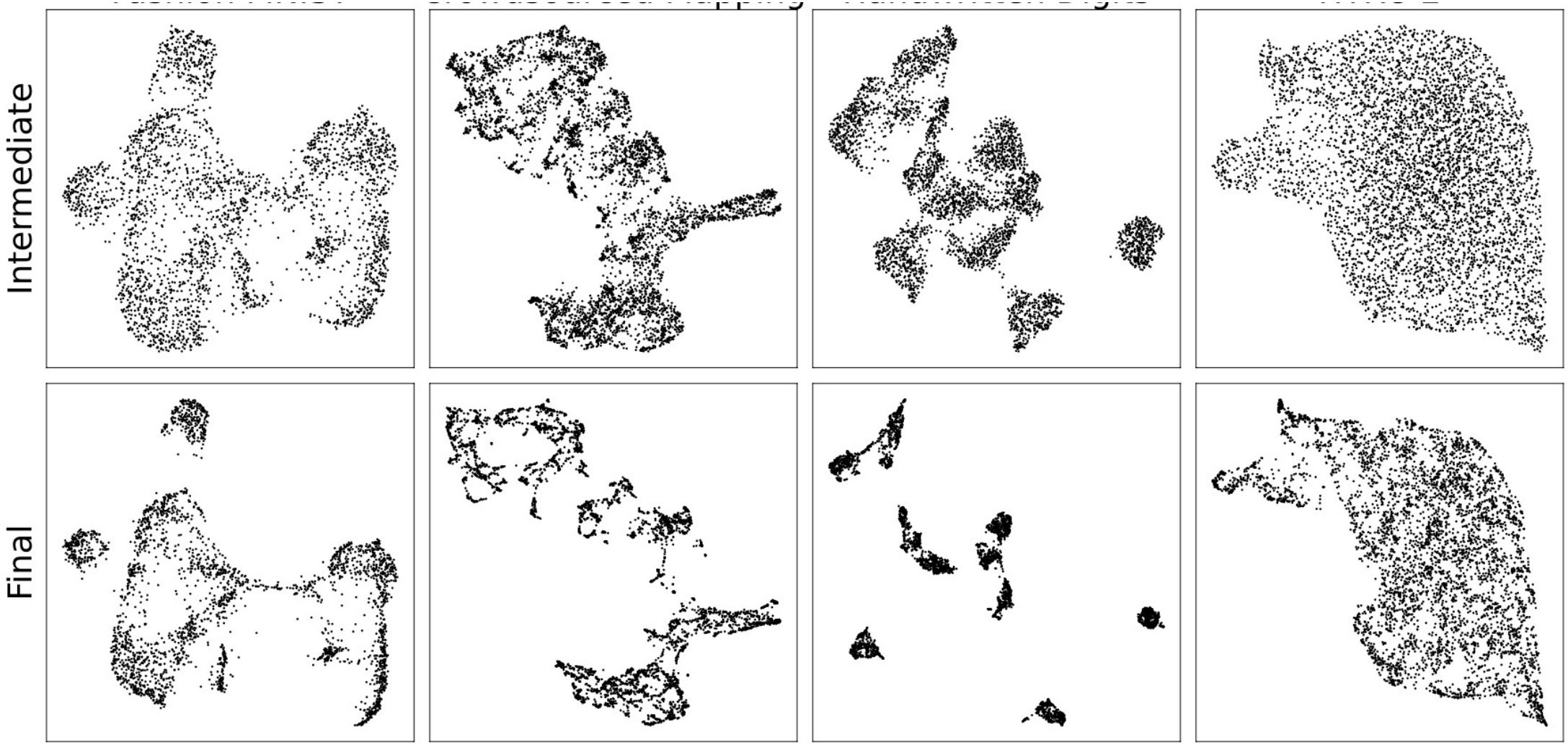
Clear Ambiguous

CLAMS IEEE VIS 2023

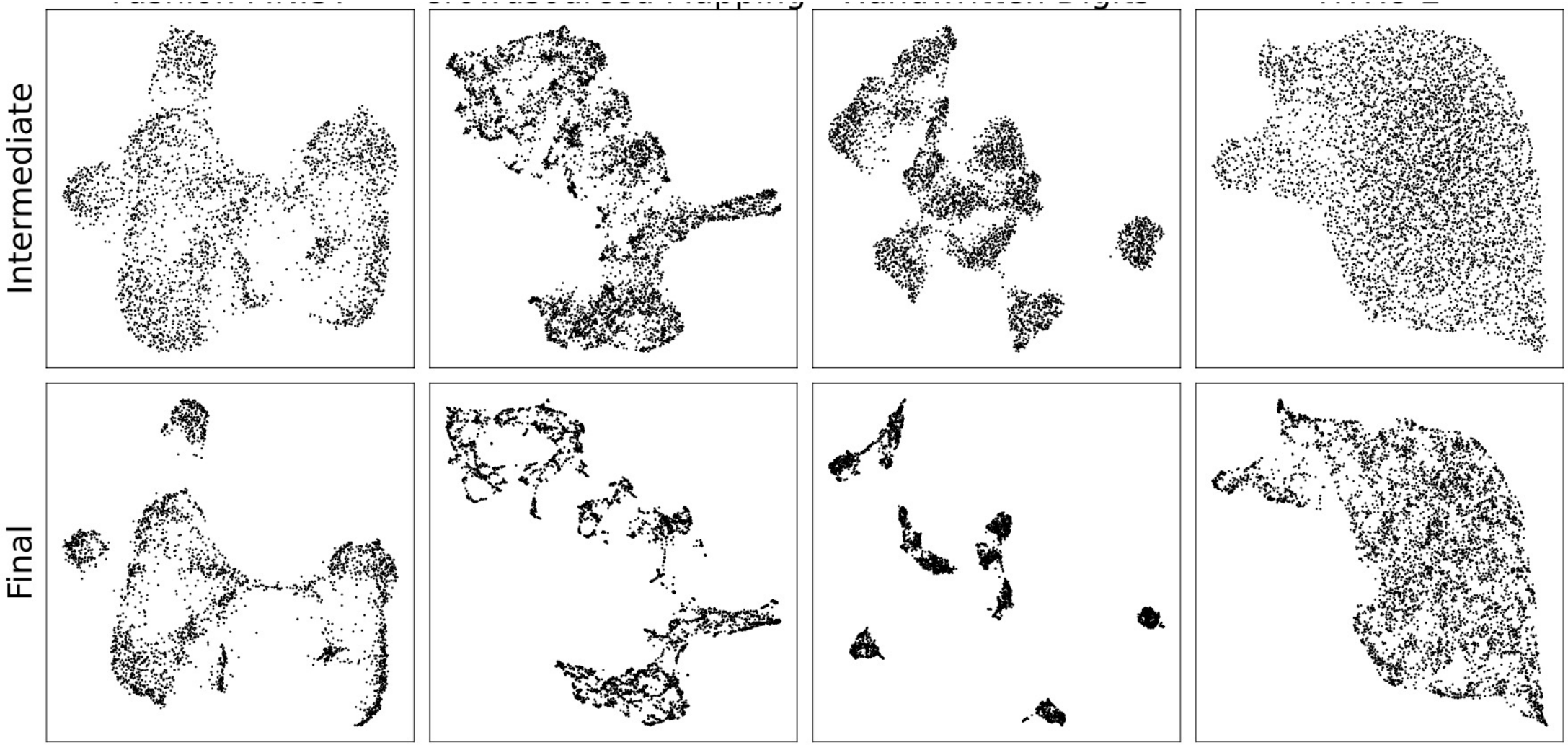
I designed **CLAMS,** a visual quality metrics that automatically quantifies the ambiguity of monochrome scatterplots



CLAMS + Dimensionality reduction



Human Factor → Dimensionality reduction





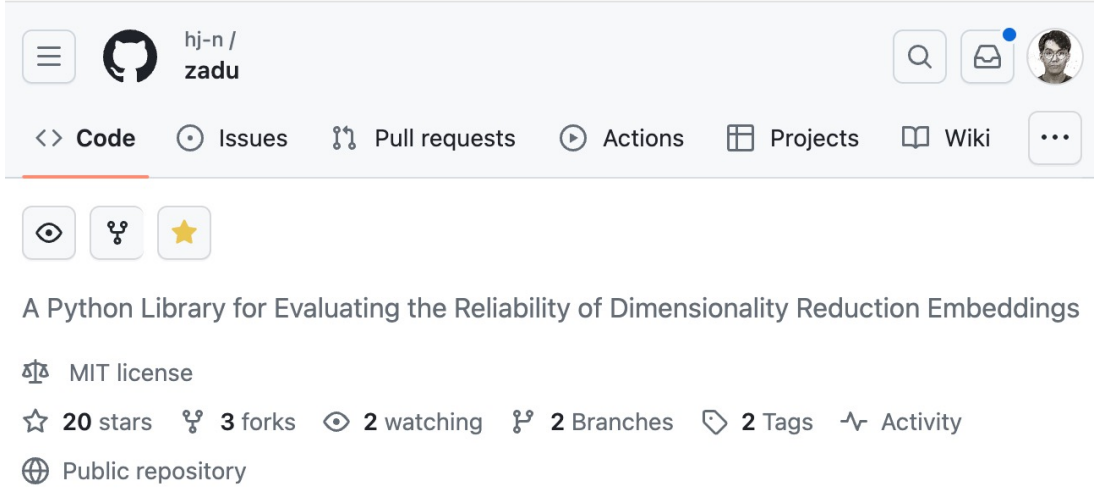
Democratizing the Findings

Making ML application reliable needs **additional effort**

How we can ***democratize our findings?***
How we can make **more practitioners to use ML more reliably?**

ZADU IEEE VIS 2023

Solution 1: Make techniques more accessible




```python
from zadu import zadu

hd, ld = load_datasets()
spec = [{
    "id"    : "tnc",
    "params": { "k": 20 },
}, {
    "id"    : "snc",
    "params": { "k": 30, "clustering_strategy": "dbscan" }
}]

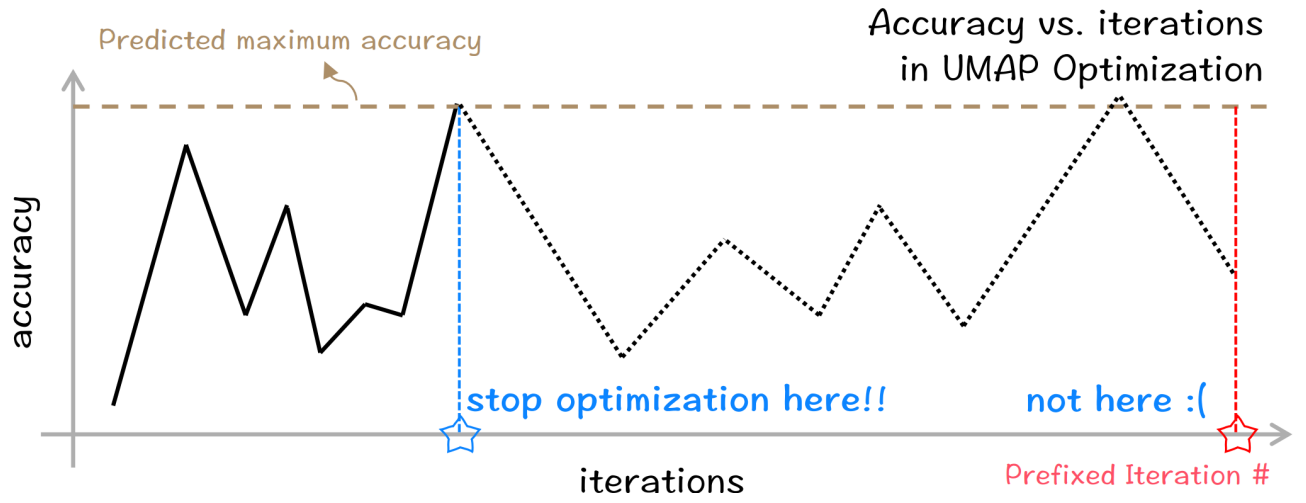
scores = zadu.ZADU(spec, hd).measure(ld)
print("T&C:", scores[0])
print("S&C:", scores[1])
```

ZADU IEEE VIS 2023

Solution 1: Make techniques more accessible

Structural Complexity *under review*, IEEE VIS 2024

Solution 2: Make techniques more efficient



ZADU IEEE VIS 2023

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Structural Complexity *under review*, IEEE VIS 2024

Solution 2: Make techniques more efficient

Future work for dissertation:

Additional Solutions??

Future Projects (Maybe after graduation)

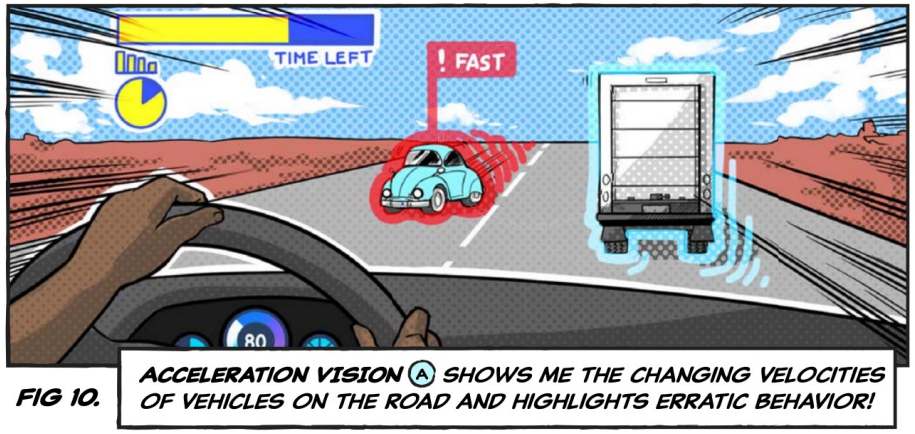
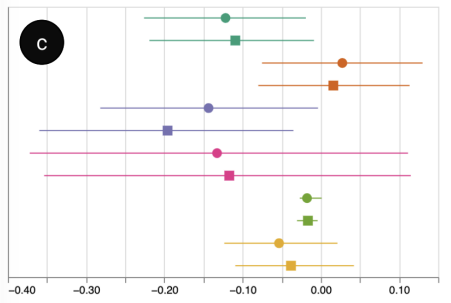


FIG 10.

Reliability problems / Evaluations
in **Immersive Analytics**



L1 caption: This layered scatter plot visualizes data about hate crimes in the UK, specifically focusing on women. The scatter plot uses points to represent the 'value' of each record, with different shapes indicating different 'models'. The points are colored and filled based on the 'group' field. A rule mark is used to indicate the range between 'min' and 'max' for each 'groupaction'. The color scheme used is 'dark2'.

Reliability problems / Evaluations
in using **LLMs for visualization**

Willett, Wesley, et al. "Perception! immersion! empowerment! superpowers as inspiration for visualization." *IEEE transactions on visualization and computer graphics* 28.1 (2021): 22-32.

Acknowledgements

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